

Quiz 1 (प्रश्नोत्तरी-1)

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1. A nanometre is one billionth of a meter (0.000 000 001 m). So, it may not come as a surprise that nanoparticles of material show different properties compared to larger particles of the same material. The surface area to volume ratio for a material or substance made of nanoparticles has a significant effect on the properties of the material. Firstly, materials composed of nanoparticles have a relatively larger surface area than the same volume of material composed of larger particles. Consider a sphere of radius “R” and prove the statement with examples.

1. एक नैनोमीटर एक मीटर का एक अरबवां हिस्सा (0.000 000 001 मीटर) होता है। इसलिए, यह आश्चर्य की बात नहीं है कि किसी पदार्थ के नैनोकण उसी पदार्थ के बड़े कणों की तुलना में भिन्न गुण प्रदर्शित करते हैं। नैनोकणों से बने पदार्थ या सामग्री के लिए पृष्ठीय क्षेत्रफल और आयतन का अनुपात पदार्थ के गुणों पर महत्वपूर्ण प्रभाव डालता है।

सबसे पहले, नैनोकणों से बने पदार्थों का पृष्ठीय क्षेत्रफल, समान आयतन वाले बड़े कणों से बने पदार्थों की तुलना में अपेक्षाकृत अधिक होता है। त्रिज्या “R” वाले एक गोले पर विचार करें और इस कथन को उदाहरण सहित सिद्ध करें।

Nanoparticles exhibit unique physical and chemical properties compared to bulk materials largely because of their massive surface area relative to their volume. As a material is divided into smaller pieces, the total surface area increases significantly while the total volume remains the same.

For example, let us consider a sphere of radius r

The surface area of the sphere will be $4\pi r^2$

The volume of the sphere = $\frac{4}{3}\pi r^3$

Therefore the surface area to the volume ratio will be $\frac{4\pi r^2}{(\frac{4}{3}\pi r^3)} = \frac{3}{r}$

It means that the surface area to volume ratio increases as the radius of the sphere decreases and vice versa. It also means that when a given volume of material is made up of smaller particles, the surface area of the material increases. Therefore, as particle size decreases, a greater proportion of the particles are found at the surface of the material. For example, a particle of size 3 nm has 50% of its particles on the surface; at 10 nm, 20% of its particles are on the surface; and at 30 nm, 5% of its particles are on the surface. Therefore, materials made of nanoparticles have a much greater surface area per unit volume ratio compared with the materials made up of bigger particles. This leads to nanoparticles being more chemically reactive. As chemical reactions occur between particles that are on the surface, a given mass of nanomaterial will be much more reactive than the same mass of material made up of large particles.

This means that materials that are inert in their bulk form are reactive when produced in their nanoparticle form.